The effect on turtles

Light pollution is a major problem for nesting sea turtles, which lay their eggs on land because the embryos breathe through the permeable shells. Females need dark, sandy beaches for their nests, and will go elsewhere if there are bright lights from beach resorts, street lights, or housing. If a whole stretch of coastline is illuminated, they may lay their eggs in inferior habitats or even deposit their eggs at sea, where their offspring will die.

Such problems may be the reason for the reduction in sea turtle populations. Scientists believe that hatchlings move toward the brightest light. In natural conditions, this will be moonlight shining on the ocean, but if there is artificial lighting inland, the hatchlings wander toward that and get run over by traffic, eaten by predators, or caught in fencing. Solutions include getting people and businesses to turn off lights at night or use “turtle-safe” lighting, which is virtually invisible to turtles.